

WM9B7-15 Artificial Intelligence & Deep Learning

Group Assessment -- Technical Implementation Report

WM9B7-15 人工智能与深度学习

小组作业 -- 技术实现完整说明

Instrument: NQ NASDAQ-100 E-mini Futures (1-min OHLCV)

Target: Automated Candlestick Pattern Recognition via TCN

Patterns: Double Bottom / Double Top / Head & Shoulders / Inverse H&S

一、问题定义 / Part 1: Problem Definition

1.1 中文说明

本项目的任务是为虚构客户 IntelliSys Ltd. 开发一套金融图形形态自动识别系统。原始数据为 NQ NASDAQ-100 期货的每分钟 OHLCV（开/高/低/收/量）历史行情，约 177 万行。

目标是让计算机像有经验的交易员一样，从价格走势中自动识别四种经典图形形态：

- Double Bottom（双底）-- 多头反转信号
- Double Top（双顶）-- 空头反转信号
- Head & Shoulders（头肩顶）-- 空头反转信号
- Inverse Head & Shoulders（头肩底）-- 多头反转信号

作业不仅要求训练模型，还要求完整的 AI 咨询报告，涵盖方法论选择、模型评估、可解释性分析以及对客户的商业建议。

1.2 English Explanation

The task is to build an automated financial chart pattern recognition system for a fictional client, IntelliSys Ltd. Raw data consists of NQ NASDAQ-100 E-mini futures in 1-minute OHLCV format (~1.77 million rows).

The objective is to train a deep learning model to automatically detect four classical chart patterns, just as an experienced trader would by visual inspection:

- Double Bottom -- bullish reversal signal
- Double Top -- bearish reversal signal
- Head & Shoulders -- bearish reversal signal
- Inverse Head & Shoulders -- bullish reversal signal

The deliverable is a complete AI consultancy report covering: methodology selection and justification, model training and evaluation, interpretability analysis, and actionable business recommendations for IntelliSys.

二、数据处理管道 / Part 2: Data Processing Pipeline

2.1 第一步：重采样（1 分钟 -> 15 分钟） / Step 1: Resampling

为什么 / Why

1 分钟数据每分钟都在小幅波动，噪声极大，形态完全看不出来。重采样到 15 分钟后，每根 K 线代表 15 分钟内的价格行为，信号更清晰，数据量从 177 万行降至约 11 万行。

1-minute data contains excessive noise, making pattern detection impossible. Resampling to 15-minute bars reduces the dataset from 1.77M to ~118K rows while preserving meaningful price structure.

方法 / Method

15-min Bar Field	Aggregation Rule
Open	First 1-min open in the 15-min window
High	Maximum of all 1-min highs
Low	Minimum of all 1-min lows
Close	Last 1-min close in the window
Volume	Sum of all 1-min volumes

2.2 第二步：Nadaraya-Watson 核回归平滑 / Step 2: NW Kernel Smoothing

为什么 / Why

即使是 15 分钟 K 线，收盘价连线仍是锯齿状。直接在上面找极值会找出几百个无意义的小波动。Nadaraya-Watson (NW) 核回归是一种非参数平滑方法，能过滤掉短期噪声，只保留主要趋势转折点。

Even 15-min closing prices remain jagged. Applying NW kernel regression produces a smooth curve that filters short-term noise and retains only the major trend reversals that define chart patterns.

原理 / Principle

```
Estimated price = SUM( K(distance) x price ) / SUM( K(distance) )
K() = Gaussian kernel: nearby points get higher weight, distant points
lower weight
Bandwidth h selected automatically via AICc (Akaike Information Criterion
corrected)
```

AICc 自动选择最优带宽，不需要手动调参，平衡平滑程度与对真实价格的拟合度。NW 核回归优于移动平均，因为它在边缘处不产生失真，且带宽由数据驱动自动确定。

AICc automatically selects the optimal bandwidth, balancing smoothness against fidelity to the original data -- no manual tuning required. This is preferred over moving averages, which suffer from edge distortion and require manual window selection.

示例图中的黑色粗线即为 NW 平滑后的价格；蓝色细线为原始 15 分钟收盘价。

In the pattern detection example figures, the black bold curve is the NW-smoothed price; the

thin blue curve is the raw 15-min close price.

三、极值点检测 / Part 3: Extrema Detection

3.1 为什么需要极值点 / Why Extrema

图形形态由特定的局部最高点（峰）和最低点（谷）组成。极值点检测是把平滑曲线转化为一串有意义的峰谷序列的关键步骤。

Chart patterns are defined by specific local peaks and troughs. Extrema detection converts the smooth price curve into a sequence of meaningful turning points.

3.2 方法：导数条件 / Method: Derivative Conditions

Condition	Mathematical Meaning
Peak (local max)	1st derivative: positive -> negative (slope turns from rise to fall)
Peak (local max)	2nd derivative < 0 (curve bends downward -- concave down)
Trough (local min)	1st derivative: negative -> positive (slope turns from fall to rise)
Trough (local min)	2nd derivative > 0 (curve bends upward -- concave up)

3.3 交替规则 / Alternation Rule

算法强制极值点必须严格交替出现：谷-峰-谷-峰...，不允许连续两个谷或两个峰。如果出现连续相同类型，保留更极端的那个（更低的谷或更高的峰），丢弃另一个。

The algorithm enforces strict alternation: trough-peak-trough-peak... Consecutive peaks or troughs are not permitted. If they occur, the more extreme one is retained (deeper trough or higher peak) and the other is discarded.

结果：整条平滑曲线被标注为一串交替的峰谷序列，例如：

Result: the entire smooth curve is annotated as an alternating sequence, e.g.:

Trough (E1) -> Peak (E2) -> Trough (E3) -> Peak (E4) -> Trough (E5) -> ...

四、形态标注（打标签） / Part 4: Pattern Labeling

4.1 什么是打标签 / What is Labeling

打标签就是对每一段 80 根 K 线的窗口数据打上 1（正样本：有形态）或 0（负样本：没有形态）。模型的学习目标就是根据这些标签，学会区分有形态和没有形态的价格序列。

Labeling assigns a binary class to each 80-bar window: 1 (positive: confirmed pattern) or 0 (negative: no pattern). The model's learning objective is to distinguish labeled pattern instances from non-pattern windows.

4.2 以 Double Bottom 为例的完整标注流程 / Full Labeling Flow (Double Bottom)

第一步：提取三元组 / Step 1: Extract Triplets

从极值序列中找所有连续的 谷-峰-谷（Trough-Peak-Trough）三元组，记为 E1、E2、E3。

Scan the extrema sequence for all consecutive Trough-Peak-Trough triplets, denoted E1 (trough), E2 (peak), E3 (trough).

第二步：几何规则检验 / Step 2: Geometry Check

Rule	Condition
Two troughs similar height	$ E1_price - E3_price / avg < 4\%$
Sufficient bounce (neckline)	$(E2_price - avg_trough) / avg_trough > 0.3\%$
Valid time separation	$10\ bars < (E3_bar - E1_bar) < 200\ bars$
Prior downtrend before E1	$(trend_start_price - E1_price) / trend_start > 0.5\%$

第三步：成交量规则 / Step 3: Volume Check

```
V(E1) > V(E3)    -- first trough volume > second trough volume
```

第一个谷底成交量大于第二个谷底成交量，说明底部成交量逐渐萎缩，是健康的反转信号。

Higher volume at the first trough than the second indicates healthy volume contraction at the support level -- a bullish signal.

第四步：突破确认（最关键） / Step 4: Breakout Confirmation (Critical)

```
for bar in range(E3_bar+1, E3_bar+20):
    if close[bar] > neckline * (1 + 0.005):      # 0.5% above neckline
        breakout_bar = bar                      # confirmed
        label = 1                               # POSITIVE sample
```

```
break
```

标签只在突破确认的那根 K 线处赋予，这是防止数据泄漏的关键设计。模型只能使用到当时已知的信息，不会看到未来价格。

The label is assigned only at the confirmed breakout bar -- this is the critical no-lookahead design. The model only uses information available at that point in time, with no access to future price data.

第五步：提取输入窗口 / Step 5: Extract Input Window

```
window = df[breakout_bar - 79 : breakout_bar + 1]    # 80 bars
# Per-window min-max normalisation (prevents price level leakage)
window = (window - window.min()) / (window.max() - window.min())
```

以突破确认 K 线为锚点，往前取 80 根 K 线作为 TCN 的输入。每个窗口独立做 min-max 归一化，消除价格绝对值的影响（防止模型记住价格水平而不是形态结构）。

The 80-bar window anchored at the breakout bar forms the TCN input. Per-window min-max normalisation eliminates absolute price levels, forcing the model to learn pattern geometry rather than memorising price ranges.

4.3 负样本：硬负样本采样 / Negative Samples: Hard Negative Mining

负样本不是随机取的，而是专门取那些看起来像但最终失败的情况，称为硬负样本：

Negatives are not random windows -- they are hard negatives: instances that resemble the pattern but fail the final confirmation:

- 失败突破：通过了几何+成交量检验，但价格最终没有突破颈线 (Failed breakout: geometry + volume passed, but price never cleared the neckline)
- 部分形态：有 E1 和 E2 但没有 E3 (Partial pattern: E1 and E2 detected but no valid E3 formed)

正负样本比例约为 1:3，强迫模型学会区分真正的形态突破与假信号。

The positive-to-negative ratio is approximately 1:3, forcing the model to discriminate genuine confirmed breakouts from near-miss patterns.

五、数据集划分 / Part 5: Train / Val / Test Split

5.1 时间切分 (Chronological Split)

全部标注窗口按锚点时间排序后，严格按时间顺序划分，不做随机打乱：

All labeled windows are sorted by anchor bar timestamp and split chronologically -- no shuffling:

Split	Purpose & Details
Train 70%	Model learns from this set. ~2018-2022 in practice.
Validation 15%	Monitors overfitting during training; used for early stopping. ~2022-2023.
Test 15%	Final evaluation only -- used once after training completes. ~2023-2024.

5.2 为什么不能随机切分 / Why Not Random Split

如果随机切分，2024 年的数据可能进入训练集，模型就见过未来数据了。在测试集上会虚假地表现很好（数据泄漏），但实盘部署时会立刻失效。金融时序数据必须严格按时间顺序切分。

Random splitting would allow future data (e.g., 2024 bars) to appear in the training set. This constitutes temporal data leakage: the model achieves artificially high test performance but fails immediately in live deployment. Chronological splitting is mandatory for financial time series.

5.3 标注是否同一套规则 / Labeling Consistency

训练集、验证集、测试集全部由同一套标注规则生成，然后再切分。这意味着 F1 分数衡量的是模型复现这套规则的能力，而不是形态是否真的有交易价值。这是基于规则标注的本质局限，在报告的局限性部分已明确说明。

All three sets are labeled by the same rule pipeline before splitting. This means F1 measures how well the model replicates the self-defined rules -- not whether detected patterns are genuinely profitable. This circular evaluation limitation is explicitly acknowledged in the Limitations section of the report.

六、数据增强 / Part 6: Data Augmentation

6.1 问题：样本极度不足 / Problem: Severely Limited Positive Samples

Double Bottom 训练集正样本只有 75 个，深度学习通常需要数万样本。直接训练的初始 F1 仅为 0.51，近似随机。

The Double Bottom training set contained only 75 positive samples. Deep learning typically requires thousands. Initial F1 was 0.51 -- near-random performance.

6.2 三种增强方法（仅对正样本，仅在训练集） / Three Augmentation Techniques (Training Positives Only)

Technique	Implementation & Rationale
Coherent OHLC noise	Same random scalar added to all 4 price columns simultaneously. Preserves Open <= High, Close >= Low constraints -- individual column noise would create invalid candlesticks.
Full Gaussian noise	Independent noise on all 5 features (OHLCV). Increases diversity across channels. std=0.012 on normalised scale.
Volume jitter	Volume column scaled by random factor in [0.88, 1.12]. Simulates realistic volume variability without distorting price structure.

6.3 结果与副作用 / Results and Side Effects

75 个正样本 x (1 + 2 副本) = 225 个正样本。训练集从 291 个增至 441 个，正负比例约 51:49。

75 positive samples x (1 + 2 copies) = 225 positives. Training set grew from 291 to 441 samples, with ~51% positive rate.

踩过的坑 / Side effect encountered: 最初设置 4 倍增强，训练集正样本达 63%，而验证集只有 16% 正样本。分布差距导致模型在验证集上崩溃，全部预测为正样本。改为 2 倍增强后解决。

Initially set to 4x augmentation: training became 63% positive while validation was only 16% positive. This distribution mismatch caused model collapse on validation (predicting all positives). Reduced to 2x copies resolved the issue.

重要原则 / Key principle: 验证集和测试集完全不做增强。实盘中只会遇到一个真实形态，不会有增强版本。对测试集增强会使评估指标失去意义。

Validation and test sets are never augmented. In practice, you encounter one real instance of a pattern -- not jittered copies. Augmenting the test set would make evaluation metrics meaningless.

七、TCN 模型架构 / Part 7: TCN Model Architecture

7.1 为什么选 TCN / Why TCN

Property	LSTM	Transformer	TCN (chosen)
Parallelisable training	No	Yes	Yes
Gradient stability	Poor	Yes	Yes
Small dataset perf.	Baseline	Overfits	Best
No future leakage	Yes	Needs masking	Yes (by design)
Memory efficient	Yes	No	Yes

对于小数据集上的固定长度窗口分类任务，TCN 是最佳选择：训练快、不需要大量数据、因果卷积天然不看未来。

For fixed-length window classification on a small dataset, TCN is optimal: fast training, low data requirements, and causal convolutions that inherently prevent future leakage.

7.2 三个核心设计 / Three Core Design Choices

因果卷积 (Causal Convolution)

每个时间步的输出只能依赖该时间步及之前的输入，永远不会使用未来数据。这是防止数据泄漏的架构级保证。

Each timestep's output depends only on that timestep and earlier inputs -- future data is structurally inaccessible. This provides an architectural guarantee against lookahead bias.

膨胀卷积 (Dilated Convolution)

```
Layer 1: dilation=1, channels=5->32 (receptive field: 3 bars)
Layer 2: dilation=2, channels=32->64 (receptive field: 7 bars)
Layer 3: dilation=4, channels=64->64 (receptive field: 15 bars)
Combined: covers the full 80-bar window with a fraction of the parameters
```

通过指数增长的膨胀率，用少量参数覆盖全部 80 根 K 线的感受野，同时保持模型轻量。

Exponentially increasing dilation rates allow the network to cover the full 80-bar window with far fewer parameters than a dense model -- essential for small dataset performance.

残差连接 (Residual Connections)

每层的输出 = 卷积结果 + 原始输入，防止深层网络梯度消失，使堆叠更多层成为可能。
Each block output = convolution result + identity shortcut. Prevents vanishing gradients and allows stacking deeper layers without degradation.

7.3 关键 Bug 修复 / Critical Bug Fix

训练初期发现模型正负样本预测概率差仅为 0.001，几乎完全没有区分能力，即使数据增强后依然如此。
Early in training, the model showed near-zero discrimination: predicted probability difference between positive and negative samples was only 0.001, even after augmentation.

根因分析 / Root Cause Analysis

信号分析发现：正样本窗口前 25%区段的价格差均值为 -0.20，后 25%区段为 +0.22，时序信号非常清晰。但模型输出无差异，说明信号在架构内部被破坏。
Feature analysis showed clear temporal signal: mean price difference in the front quarter of positive windows was -0.20, rear quarter was +0.22. The signal existed in the data but was destroyed inside the model.

Code	Effect
BEFORE (bug): <code>x = x.mean(dim=-1)</code>	Global Average Pooling averages all 80 timesteps. Front-quarter low values (-0.20) averaged with rear-quarter high values (+0.22) cancel to ~0. Signal destroyed.
AFTER (fix): <code>x = x[:, :, -1]</code>	Last timestep readout. In a causal TCN, the final timestep has already encoded the full 80-bar history through stacked dilated convolutions. Full temporal signal preserved.

效果 / Impact: F1 从 0.51 提升到 0.68 -- 一行代码修复，原理清晰。
F1 improved from 0.51 to 0.68 with a single-line fix grounded in clear theoretical reasoning.

八、训练过程 / Part 8: Training Process

8.1 超参数 / Hyperparameters

Hyperparameter	Value & Rationale
Optimiser	AdamW -- adaptive learning rate with weight decay decoupling

Learning rate	2e-4 -- stable convergence on this scale after tuning
Weight decay	1e-3 -- stronger L2 regularisation for small dataset
Class weights	[1.0, 1.5] -- slight positive class boost after augmentation
Max epochs	60 -- sufficient for convergence with early stopping
Early stopping patience	10 -- stop if val F1 doesn't improve for 10 epochs
Seed	42 -- full reproducibility

8.2 Early Stopping (早停)

每个 Epoch 结束后在验证集上计算 F1。若连续 10 个 Epoch F1 没有提升，训练自动停止，使用历史最好的验证集 F1 对应的模型权重。

After each epoch, macro F1 is computed on the validation set. If no improvement occurs for 10 consecutive epochs, training stops and the best checkpoint is restored. This prevents overfitting without manual intervention.

8.3 训练曲线怎么读 / How to Read Training Curves

Curve	What to Look For
Train Loss	Should decrease steadily, then plateau. Rapid decrease = learning; plateau = convergence.
Val Loss	Should track Train Loss. If Val rises while Train falls, overfitting has begun.
Val F1	Should increase overall, showing improving generalisation. Early stopping fires at the peak.

九、模型评估 / Part 9: Model Evaluation

9.1 为什么用 F1 而不是准确率 / Why F1 over Accuracy

数据集正负样本不平衡（约 1:3）。如果模型全部预测为负样本，准确率仍有 75%，但对业务毫无意义。F1 分数同时考虑了精确率和召回率，是不平衡分类的正确指标。

The dataset is imbalanced (~1:3 positive to negative). A trivial classifier predicting all negatives achieves 75% accuracy -- a meaningless metric here. F1 combines precision and recall, providing an honest measure for imbalanced classification.

9.2 混淆矩阵各格含义 / Confusion Matrix Interpretation

Cell	Meaning (ZH)	Meaning (EN)	Business Impact
True Positive	正确识别出形态	Correctly identified pattern	Successful trade entry
True Negative	正确排除假信号	Correctly rejected non-pattern	Avoided losing trade
False Positive	把假信号误判为形态	False alarm	Wrong entry, potential loss
False Negative	漏掉真实形态	Missed pattern	Missed opportunity profit

9.3 最终结果 / Final Test Results

Pattern	Macro F1	Accuracy	Assessment
Double Top	0.973	97.9%	Excellent -- production ready
Inverse H&S	0.856	86.5%	Excellent -- production ready
Double Bottom	0.756	77.8%	Good -- monitor in production
Head & Shoulders	0.306	44.2%	Failed -- documented as negative finding

H&S 失败的原因 / Why H&S Failed: NQ 15 分钟数据上严格意义的头肩顶出现频率极低，原始严格参数只扫出 14 个正样本。放松参数到 7%对称容差并关闭成交量要求后，样本扩至 256 个，但标注质量下降，模型连放松后的规则都学不好。这作为负面发现在报告中正面说明，而非隐瞒。

Strict H&S parameters yielded only 14 positive samples on NQ 15-min data. Relaxing symmetry

tolerance to 7% and disabling volume requirements increased samples to 256, but at the cost of label quality. The model could not learn consistent features from these heterogeneous samples. This is documented as a meaningful negative finding rather than omitted.

十、Grad-CAM 可解释性分析 / Part 10: Grad-CAM Interpretability

10.1 为什么需要可解释性 / Why Interpretability

深度学习的黑盒问题在金融领域尤其严重。如果模型偶然学到了某个无关的噪声规律，在真实市场上会立刻失效。Grad-CAM 将模型决策时关注的时间步可视化出来，让人类可以验证模型是否学到了真实的形态结构。

The black-box nature of deep learning is especially problematic in finance. If the model learned a spurious noise pattern, it will fail immediately in live trading. Grad-CAM visualises which timesteps drive the model's decision, enabling human verification that the model learned genuine pattern structure rather than artefacts.

这也对应了 MiFID II 算法交易监管要求中的可解释性义务，对 IntelliSys 的合规部门具有直接价值。

This directly addresses explainability obligations under MiFID II algorithmic trading regulations -- directly relevant to IntelliSys's compliance requirements.

10.2 工作原理 / How It Works

- 正向传播：输入窗口通过网络得到预测结果 (Forward pass: input window produces prediction)
- 反向传播：计算预测类别相对于最后一个 TCN 残差块输出的梯度 (Backward pass: compute gradients of predicted class w.r.t. last TCN block activations)
- 注意力图：梯度绝对值越大的时间步 = 对决策贡献越大 = 模型在看这里 (Attention map: timesteps with larger gradient magnitude contributed more to the decision)
- 归一化到[0,1]，可视化为每个时间步上方的粉色柱状图 (Normalised to [0,1] and visualised as pink bars above each timestep)

10.3 结果解读 / Results Interpretation

Attention Location	Interpretation
Near E1 (first trough)	Model identifies the first support level -- anchor of the W-shape
Near E3 (second trough)	Model confirms the second test of support -- key

	confirmation point
Window tail (bar 70-79)	Model registers the breakout -- the label anchor
Middle section (bar 15-50)	Low attention -- the recovery phase is not informative for the decision

注意力高峰与人类交易员看图时关注的关键点完全一致：两个谷底 + 最终突破。这证明模型学到的是真实的形态结构，而不是数据集中的随机噪声。

Attention peaks align precisely with the decision points a human trader would focus on: the two troughs and the final breakout. This constitutes strong evidence that the model learned genuine pattern geometry, not spurious dataset correlations.

十一、回测验证 / Part 11: Backtest Validation

11.1 为什么需要回测 / Why Backtest

分类指标（F1）只衡量模型复现标注规则的能力，不能证明识别出的形态有实际盈利价值。回测是唯一一个完全独立于标注规则的验证维度。

Classification metrics (F1) only measure rule-replication ability. A backtest is the only validation dimension fully independent of the labeling rules -- it tests whether the detected patterns actually generate positive expected returns.

11.2 回测逻辑 / Backtest Logic

Parameter	Setting
Entry	Buy at close of confirmed breakout bar
Exit	Close position after 8 hours (32 x 15-min bars)
Baseline	Random entry at same holding period, same number of trades
Metric	Win rate (% trades with positive return) and total return

11.3 结果 / Results

Metric	Double Bottom Signal vs Random Baseline
Win rate (8h)	62.5% (model) vs ~50% (random)
Total return	+11.6% (model) vs +0.6% (random)

Conclusion

Model signals demonstrate positive directional edge over random entry

11.4 局限性 / Limitations

- 不含手续费和滑点 (No transaction costs or slippage)
 - 固定持仓时间，无止损/止盈 (Fixed holding period, no stop-loss or take-profit)
 - 样本量较少，统计显著性有限 (Small sample size -- limited statistical significance)
 - 单一品种验证，泛化能力未知 (Single instrument -- generalisability unknown)
-

十二、完整流程总结 / Part 12: End-to-End Summary

12.1 流程图 / Pipeline Diagram

```
Raw 1-min OHLCV (1.77M rows)
  |-- Resample to 15-min (~118K bars)
  |-- Nadaraya-Watson smoothing (AICc bandwidth selection)
  |-- Derivative-based extrema detection (peaks & troughs, alternation
enforced)
  |-- Pattern geometry + volume + breakout confirmation rules
  |-- Labeled windows (80 bars, anchor at breakout bar, per-window
normalisation)
  |-- Chronological split: Train 70% | Val 15% | Test 15%
  |-- Data augmentation (training positives only: 3 techniques x 2
copies)
  |-- TCN training (causal dilated conv + residual + last-timestep
readout)
  |-- Evaluation: confusion matrix + F1 (test set, used once)
  |-- Grad-CAM interpretability (attention maps on correctly predicted
positives)
  |-- Backtest (entry at breakout bar, 8h hold, vs random baseline)
```

12.2 所有模式汇总 / All Patterns Summary

Pattern / 形态	Macro F1	Test Accuracy	Status
Double Top / 双顶	0.973	97.9%	Production ready
Inverse H&S / 头肩底	0.856	86.5%	Production ready
Double Bottom / 双底	0.756	77.8%	Good (monitor)
Head & Shoulders / 头肩顶	0.306	44.2%	Failed -- negative finding

12.3 关键发现 / Key Findings

- GAP Bug 修复是性能提升最大的单一改动 (F1: 0.51->0.68) -- Global Average Pooling destroyed the temporal signal that defines Double Bottom patterns. Fix: last-timestep readout.
- 数据增强有效但需控制倍数 -- 2x copies gave balanced training; 4x caused distribution mismatch and model collapse on validation.
- H&S 失败是有意义的负面发现 -- NQ 15-min volume decay patterns are unstable; the pattern definition itself constrains sample availability in a way that cannot be resolved by parameter relaxation alone.

- Grad-CAM 证明模型学到了真实结构 -- attention peaks align with E1 trough, E3 trough, and breakout bar: the three structural anchors of the Double Bottom pattern.
- 回测验证实际盈利能力 -- 62.5% win rate vs ~50% random baseline demonstrates genuine directional edge, independent of labeling rules.

12.4 局限性与未来工作 / Limitations & Future Work

Limitation / 局限性	Mitigation / 改进方向
Self-defined labels (circular eval)	Independent annotation by professional traders; external signal validation
Small dataset (<300 positives)	Multi-instrument multi-year data; transfer learning from larger corpus
H&S failure on NQ 15-min	Longer timeframe (1h/4h); unsupervised clustering to verify pattern existence
Simple backtest (no costs)	Realistic cost model; Sharpe ratio; walk-forward validation
Single instrument (NQ only)	Cross-validate on ES, CL, EURUSD before deployment